CUSTOMER CHURN PREDICTION

Machine Learning Project Report

Project objective

Develop a customer churn prediction prototype that analyses customer attributes, compares machine-learning models, estimates churn risk, and supports retention planning.

Project scope

Data preparation • Exploratory analysis • Feature engineering • Model comparison • Threshold and cost analysis • Batch prediction • Business insights • Drift-monitoring prototype

Project dataset summary	Value
Customer records after cleaning	440,832
Churned customers	249,999 (56.71%)
Non-churned customers	190,833 (43.29%)
Selected prediction pipeline	Random Forest

Prepared as a prototype using the supplied customer churn dataset. Student name, institution, internship provider and submission date can be added before submission.

1. Methodology and Implementation

The source dataset contained customer demographics, tenure, usage frequency, support calls, payment delay, subscription type, contract length, total spend, last interaction and a churn label. An empty row was removed; the cleaned dataset contained 440,832 records with no remaining missing values or duplicate customer IDs.

Feature engineering

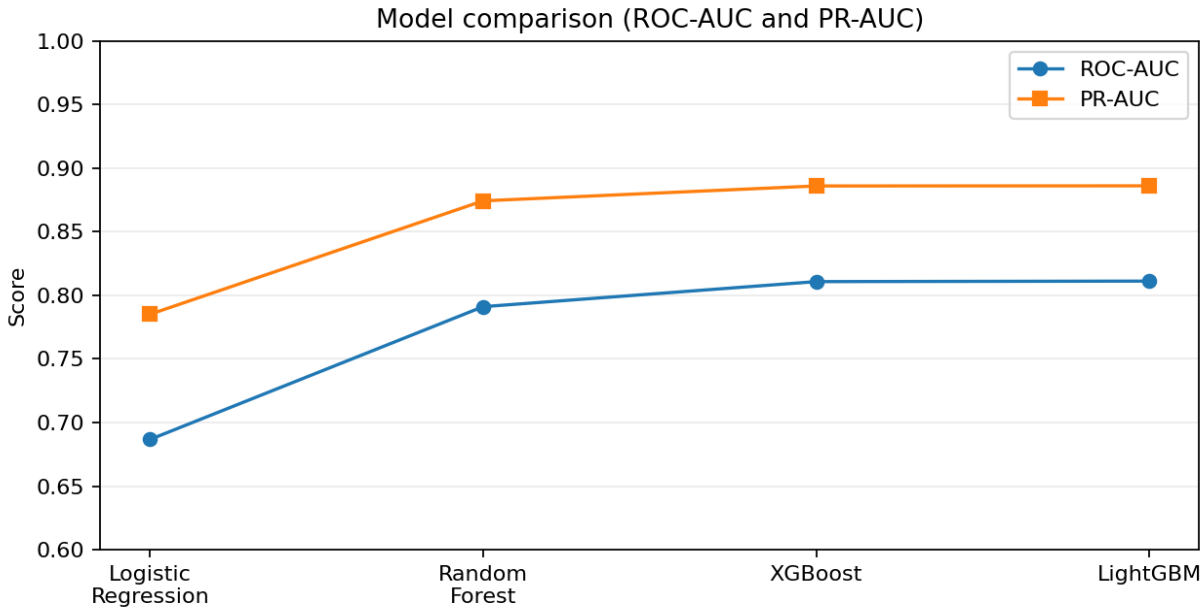
Engineered variables included Usage_Intensity (usage frequency divided by tenure) and Inactivity_Flag (last interaction of at least 20 days). Additional derived variables were explored, but suspicious features based on support calls, payment delay, total spend and contract length were excluded from the strict model feature set.

Final strict feature set

Age, Tenure, Usage Frequency, Last Interaction, Usage_Intensity, Inactivity_Flag, Gender_Male, Subscription Type_Premium and Subscription Type_Standard.

Models compared

Model	Accuracy	ROC-AUC	PR-AUC
Logistic Regression	61.93%	0.6867	0.7851
Random Forest	71.78%	0.7911	0.8743
XGBoost	74.92%	0.8108	0.8860
LightGBM	74.90%	0.8112	0.8861



XGBoost and LightGBM performed very similarly. Random Forest was retained for the current saved prediction pipeline and threshold/cost demonstration; this does not mean it was the top model on every comparison metric.

2. Evaluation and Business Findings

The saved Random Forest pipeline uses a provisional probability threshold of 0.30. On the held-out batch prediction file, churn precision was 0.64 and churn recall was 0.86; overall accuracy at this threshold was approximately 64%. The ROC-AUC was 0.7911.

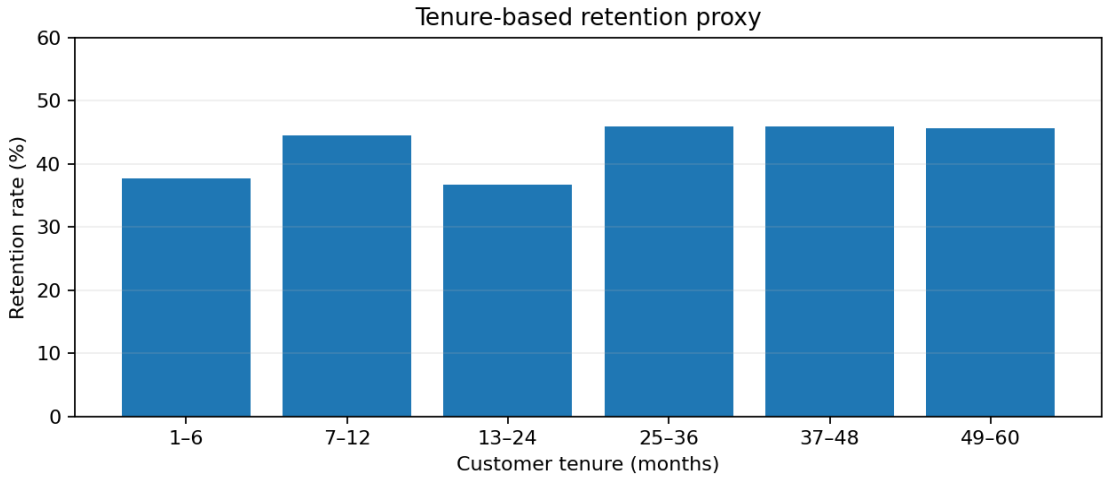
Threshold and illustrative cost analysis

Thresholds from 0.30 to 0.70 were compared. Under an illustrative cost assumption of 1 unit for a false positive and 5 units for a false negative, threshold 0.30 had the lowest estimated total cost (60,229 units) among the tested thresholds. These are not real monetary costs.

Business patterns observed

Segment	Observed churn rate
Monthly contracts	100.00%
Payment delay 21–30 days	100.00%
Support calls 5–10	99.08%
Last interaction 16–30 days	66.50%

Tenure-based retention proxy

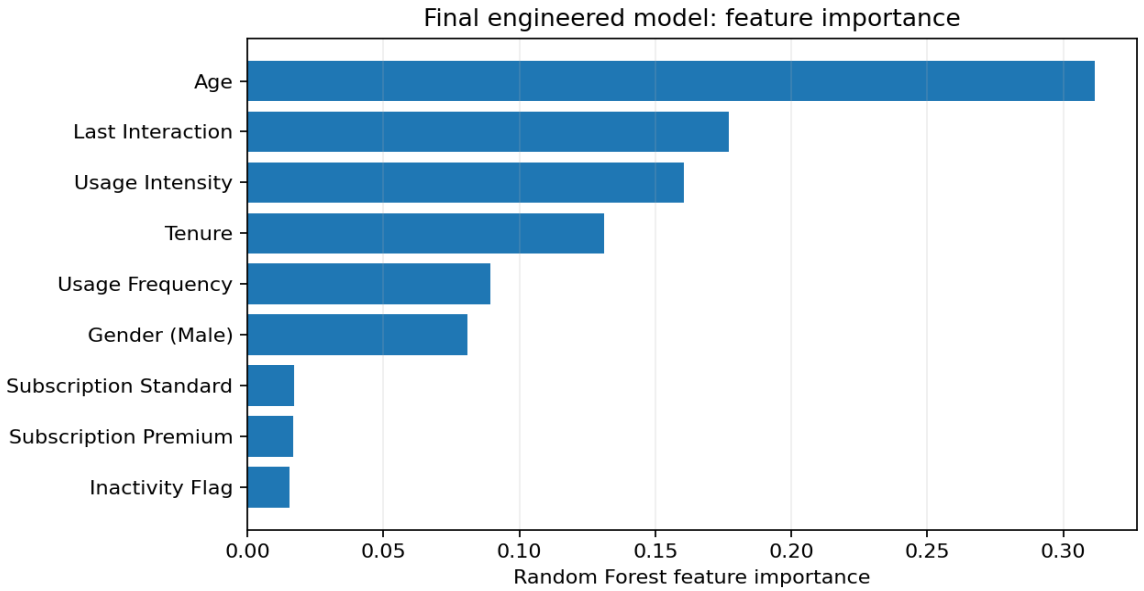


These are cross-sectional tenure-group retention proportions, not longitudinal cohort retention curves. The dataset does not contain verified signup dates or repeated customer observations.

3. Explainability, Deployment and Limitations

Explainability

Feature importance from the final engineered Random Forest ranked Age, Last Interaction, Usage_Intensity and Tenure among the strongest features. Approximate TreeSHAP explanations were generated for a 20-row sample from the final model and saved as summary and bar plots. The small sample and approximate calculation should be disclosed when interpreting the plots.



Prototype deployment components

The project saves a trained model bundle, generates batch predictions, and scores a small synthetic new-customer file. A drift-monitoring script compares a small incoming sample with the reference data. The five-row sample is only a demonstration and is not sufficient for reliable drift conclusions.

Limitations and future work

The dataset is cross-sectional and has no verified event timestamps, signup dates or censoring information; therefore genuine time-based cohorts and Kaplan–Meier survival analysis cannot be validated. Perfect-churn patterns in some segments may reflect dataset-generation rules; their cause is unknown. Airflow scheduling, MLflow tracking, CRM/reverse-ETL integration, real-time monitoring and automated retraining are proposed future integrations, not implemented services. Model results require validation on independent, representative business data.

Conclusion

The project delivers an end-to-end churn prediction prototype covering data preparation, model comparison, risk scoring, explainability, business analysis and preliminary monitoring. Its results are promising within this dataset but should not be interpreted as guaranteed real-world predictive performance.

Dataset source: MatanKriel, “Matan-Kriel-Customer-Churn-dataset”, Hugging Face.
https://huggingface.co/datasets/MatanKriel/Matan-Kriel-Customer-Churn-dataset